05-11 第1篇

阅读理解（说明文） How Ants Build Their Huge Underground Cities

① At a picnic, it is easy to notice ants walking in long lines, carrying a crumb of bread or a piece of rice. Most children wave a quick hand and forget about them. But if you could shrink to the size of a coin and follow the line, you would find something amazing — a whole **underground city**, sometimes bigger than your classroom, built by tiny workers with no map and no boss telling them what to do.

② An ant colony is made up of three main groups. The **queen** is usually the largest. Her only job is to lay eggs, sometimes thousands each week. Around her live **worker ants**, who do almost everything else: digging tunnels, finding food, feeding babies, and cleaning rooms. A few **male ants** appear only at certain times of the year, to help the queen produce the next generation.

③ The city begins with a single decision. A young queen, after flying away from her old nest, finds a safe spot of soft soil. With her strong jaws, she digs a small room, drops her wings — she no longer needs them — and starts laying her first eggs. Within a few weeks, the first workers hatch and begin to **widen** the tunnels.

④ A mature ant nest has many kinds of rooms: nurseries for eggs and baby ants; cool rooms for food storage; a warm "royal room" for the queen; and even small "rubbish rooms" where dead ants and old food are placed. Tunnels connect all these rooms like streets in a city. Some species even build small air vents, so that deep rooms get fresh air on hot days.

⑤ How do so many ants work together without a boss shouting orders? The secret is **chemical messages**. When a worker finds food, she walks home while leaving a thin chemical trail on the ground. Other ants follow the trail and find the food too. When the food is gone, the ants stop leaving the chemical, and the trail slowly fades away. In the same way, ants use chemicals to mark danger, to call for help, and to recognise each other as "family".

⑥ Scientists who study these tiny cities have learned ideas that help humans too. Modern traffic engineers now copy the way ants avoid jams, and computer experts design delivery systems based on ant chemical trails. So next time a thin line of ants crosses your floor, do not rush to chase them away. You are looking at one of nature's oldest and cleverest classrooms — built from sand, saliva and teamwork.

根据短文内容，选择最佳答案。

1. How many main kinds of ants live in a colony?

A. One.
B. Two.
C. Three — queen, workers and males.
D. Five.

2. What does a young queen do after flying away from her old nest?

A. She flies back to the old queen.
B. She finds a safe spot, digs a small room, drops her wings and starts laying eggs.
C. She looks for a new boss.
D. She hides inside a tree for the rest of her life.

3. Why are there small air vents in some ant nests?

A. To let in more ants.
B. So that deep rooms get fresh air on hot days.
C. To let the rain come inside.
D. To make music.

4. How do ants share information about food?

A. They scream loudly.
B. They wave their legs in the air.
C. They leave a chemical trail that others follow.
D. They use mobile phones.

5. What does the writer MOST want to show?

A. Ants are boring and useless.
B. Ants build amazing cities by teamwork and chemical messages, and humans can learn from them.
C. Ants can fly all year.
D. Ants hate humans.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第②段第1–2句 "three main groups. The queen... worker ants... male ants..."
【解析】一个蚁群有三种主要蚂蚁。A/B/D 均不符合。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "finds a safe spot of soft soil... digs a small room, drops her wings... starts laying her first eggs."
【解析】年轻蚁后找软土、挖小房、脱翅、产卵。A/C/D 均与原文不符。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段末 "small air vents, so that deep rooms get fresh air on hot days."
【解析】通风孔为深处房间提供新鲜空气。A/C/D 均无据。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑤段第2句 "she walks home while leaving a thin chemical trail..."
【解析】蚂蚁留下化学痕迹让同伴跟随。A/B/D 均不符。故选 C。

5.
【答案】B
【题型】主旨大意/写作意图题
【定位】第⑥段 "nature's oldest and cleverest classrooms" 以及"人类向蚂蚁学习"
【解析】作者赞叹蚂蚁的协作与启发意义。A/C/D 均与原文相反。故选 B。

词汇注释（超纲词）

- **colony** [n.] （蚁、蜂）群体
- **jaw** [n.] 颚；下巴
- **tunnel** [n.] 隧道；通道
- **chemical trail** [n. phr.] 化学痕迹；信息素路径